

Statistical Mispricing and Arbitrage Detection in Bitcoin Prediction Markets: A Black–Scholes Approach to Kalshi vs. Deribit

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Abstract

This project investigates whether statistical mispricing exists between short-dated Bitcoin threshold contracts on Kalshi (a regulated US prediction-market exchange) and volatility priced into BTC options on Deribit. A digital-option pricing model was derived from first principles under the Black–Scholes framework, implemented without third-party statistics libraries, and validated against one week of real, resolved Kalshi KXBTCD markets matched with contemporaneous Deribit volatility data. The model showed genuine, statistically significant discriminative power and a persistent, significant gap between Kalshi-implied and Deribit-implied volatility ($p \approx 5.6 \times 10^{-9}$), alongside a systematic calibration bias plausibly attributable to a volatility term-structure mismatch. A simulated trading book built on this signal is presented, alongside an honest account of why a naive position-sizing rule produced a fragile, outlier-dependent result, and how a disciplined rule preserved the edge with a materially healthier risk profile.

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1 Introduction and Thesis

Kalshi is a CFTC-regulated exchange offering binary contracts on real-world events, including short-dated Bitcoin threshold markets (KXBTC) that resolve hourly against CF Benchmarks' Bitcoin Real-Time Index. Because each contract is structurally a *binary digital option* on BTC price, its fair value can, in principle, be derived from the same option-pricing theory used for any digital option: given a spot price, a strike, a time to expiry, and a volatility estimate, Black–Scholes theory gives an exact fair probability of finishing above the strike.

This creates a natural test: if an independent, high-quality estimate of BTC volatility is available from a deep, liquid options market (Deribit), it can be fed into a self-built pricing model and compared against Kalshi's own market price for the same event. A persistent, statistically significant divergence between the two would constitute evidence of genuine statistical mispricing – not because the two exchanges are pricing the exact same instrument, but because they are pricing correlated bets on the same underlying stochastic process, and should therefore imply similar volatility if both are efficient.

The project deliberately avoided using pre-built statistical or options-pricing libraries (e.g. `scipy.stats`) for any of the core mathematics, in favour of deriving and implementing the normal CDF, its inverse, the digital-option pricing formula, the implied-volatility solver, and the one-sample *t*-test entirely from first principles. This was a deliberate pedagogical choice: the goal was not merely to obtain a result, but to build a from-scratch, fully understood quantitative pipeline that could be extended to other markets in future work.

2 System Architecture

The system is organised into three layers, corresponding to the stages of the analysis:

1. **Data layer** (`data/`) – two REST clients, `kalshi_client.py` and `deribit_client.py`, both built with shared exponential-backoff retry logic against rate-limited public endpoints, and full cursor-based pagination.
2. **Modelling layer** (`models/fair_value.py`) – the pricing mathematics described in Section 3: the digital-option pricer, its algebraic inverse (implied volatility), and the underlying special functions each depends on.
3. **Backtesting layer** (`backtest/`) – a historical data collector that reconstructs, for each resolved Kalshi market, four snapshots in time (60, 30, 10, and 2 minutes before close) with matched spot price, Kalshi quote, and Deribit volatility; a calibration script implementing quantile-based calibration testing and a hand-rolled hypothesis test; and a simulated trading book.

2.1 Data Collection Methodology

For each resolved hourly KXBTC market, the collector:

1. Identifies the strike closest to BTC's historical spot price at the hour's open, using Deribit's BTC-PERPETUAL index as the spot reference (approximating an at-the-money contract).
2. At each of four checkpoints before close, independently re-fetches spot price, Kalshi's live bid/ask for that specific contract, and Deribit's 30-day volatility index (DVOL), each via a nearest-timestamp lookup against the respective exchange's minute-level historical data.
3. Computes the model's fair probability using the volatility fetched at that checkpoint, and records it alongside the actual Kalshi price and the market's eventual, real settlement outcome.

This process was run across a full 7-day window, yielding **655 checkpoint observations** across 164 distinct hourly markets, each anchored to a genuine, resolved outcome.

3 Mathematical Derivations

3.1 The Standard Normal CDF

A Kalshi KXBTC contract pays \$1 if BTC's spot price at expiry is above a strike K , and \$0 otherwise – structurally identical to a cash-or-nothing digital call option. Under the standard Black–Scholes assumption that $\ln S_T$ is normally distributed, the fair probability of finishing above the strike is

$$P(S_T > K) = N(d_2), \quad d_2 = \frac{\ln(S/K) + (r - \frac{1}{2}\sigma^2) T}{\sigma\sqrt{T}} \quad (1)$$

where S is the current spot price, K the strike, r the risk-free rate, σ the annualised volatility, T the time to expiry in years, and $N(\cdot)$ the standard normal cumulative distribution function.

$N(x)$ has no elementary closed form, but is related to the error function erf by the identity

$$N(x) = \frac{1}{2} \left(1 + \text{erf}\left(\frac{x}{\sqrt{2}}\right) \right) \quad (2)$$

which was implemented directly using Python's `math.erf`, itself a standard-library implementation of the error function.

3.2 The Digital Option Pricer

Equations (1)–(2) combine into the core forward-pricing function: given (S, K, T, σ, r) , compute d_2 , then $N(d_2)$. Two degenerate cases were guarded against explicitly: $T \leq 0$ (a market already past expiry) and $\sigma \leq 0$ (volatility must be strictly positive, since it appears in a denominator in d_2), both of which raise an explicit exception rather than propagate a silent NaN or a divide-by-zero.

3.3 The Inverse Normal CDF

To invert Equation (1) – i.e. to back out an implied volatility from an observed market price – an inverse of $N(\cdot)$ is required first, since $d_2 = N^{-1}(p)$. Unlike $N(x)$, $N^{-1}(p)$ has no useful closed form even in terms of erf^{-1} for computational purposes, and was instead approximated using **Peter Acklam's rational approximation algorithm**, which fits three separate low-order rational polynomials over three regions of $p \in (0, 1)$:

$$N^{-1}(p) \approx \begin{cases} \frac{(((c_0q + c_1)q + c_2)q + c_3)q + c_4)q + c_5}{(((d_0q + d_1)q + d_2)q + d_3)q + 1}, & p < p_{\text{low}} \\ \frac{(((a_0r + a_1)r + a_2)r + a_3)r + a_4)r + a_5)q}{(((b_0r + b_1)r + b_2)r + b_3)r + b_4)r + 1}, & p_{\text{low}} \leq p \leq p_{\text{high}} \\ - \left[\text{lower-tail formula with } q = \sqrt{-2 \ln(1-p)} \right], & p > p_{\text{high}} \end{cases} \quad (3)$$

where, for the central region, $q = p - 0.5$, $r = q^2$; for the lower tail, $q = \sqrt{-2 \ln p}$; and $p_{\text{low}} = 0.02425$, $p_{\text{high}} = 1 - p_{\text{low}}$. The tail regions use a variable transformation ($q = \sqrt{-2 \ln p}$) specifically because the required output diverges as $p \rightarrow 0$ or $p \rightarrow 1$, and no single low-order polynomial fits both the well-behaved central region and the rapidly diverging tails simultaneously.

Verification. The implementation was validated by a round-trip test: for a known x , $N^{-1}(N(x))$ was confirmed to recover x to within 10^{-8} across the full domain, including both tail regions and the central region boundary.

3.4 Implied Volatility: Inverting the Pricer

With $N^{-1}(\cdot)$ available, d_2 can be recovered directly from an observed market probability p : $d_2 = N^{-1}(p)$. Substituting into the definition of d_2 and rearranging as a polynomial in σ yields a genuine **quadratic equation**:

$$\begin{aligned} d_2 \sigma \sqrt{T} &= \ln(S/K) + (r - \tfrac{1}{2}\sigma^2) T \\ \tfrac{1}{2}T \sigma^2 + d_2 \sqrt{T} \sigma - (\ln(S/K) + rT) &= 0 \end{aligned} \quad (4)$$

which is of the form $A\sigma^2 + B\sigma + C = 0$ with

$$A = \tfrac{1}{2}T, \quad B = d_2 \sqrt{T}, \quad C = -\ln(S/K) - rT$$

Solved via the quadratic formula, $\sigma = \frac{-B + \sqrt{B^2 - 4AC}}{2A}$, taking only the positive root, since volatility cannot be negative. Two further guards were added: the discriminant $B^2 - 4AC$ must be non-negative (otherwise no real volatility explains the observed price, given the other inputs), and the resulting σ must itself be strictly positive.

Verification. A full round-trip was confirmed: for an arbitrary $\sigma = 0.3$, the forward pricer produced a probability which, fed back into the implied-volatility solver, recovered $\sigma = 0.300000000200\dots$ – correct to nine decimal places, and independently re-confirmed at every one of the 655 checkpoints in the final historical dataset via an in-line assertion (Section 4).

3.5 The One-Sample t -Test

To test whether the mean gap between Kalshi-implied and Deribit-implied volatility differs significantly from zero, a one-sample t -test was hand-implemented. Given a sample of n gaps $x_1, \dots, x_n$:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i \quad (\text{sample mean}) \quad (5)$$

$$s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2} \quad (\text{sample standard deviation, Bessel-corrected}) \quad (6)$$

$$\text{SE} = \frac{s}{\sqrt{n}} \quad (\text{standard error of the mean}) \quad (7)$$

$$t = \frac{\bar{x}}{\text{SE}} \quad (\text{test statistic}) \quad (8)$$

With n in the hundreds, the t -distribution converges closely to the standard normal, so rather than implementing the (considerably heavier) incomplete beta function required for an exact Student's- t CDF, the existing $N(\cdot)$ from Equation (2) was reused as a large-sample approximation for the two-tailed p -value:

$$p = 2(1 - N(|t|)) \quad (9)$$

4 Findings

4.1 Forward-View Calibration

Model-predicted probabilities were sorted and split into 8 equal-population quantile buckets (≈ 81 – 88 observations each). Within each bucket, the average predicted probability was compared against the actual resolution rate.

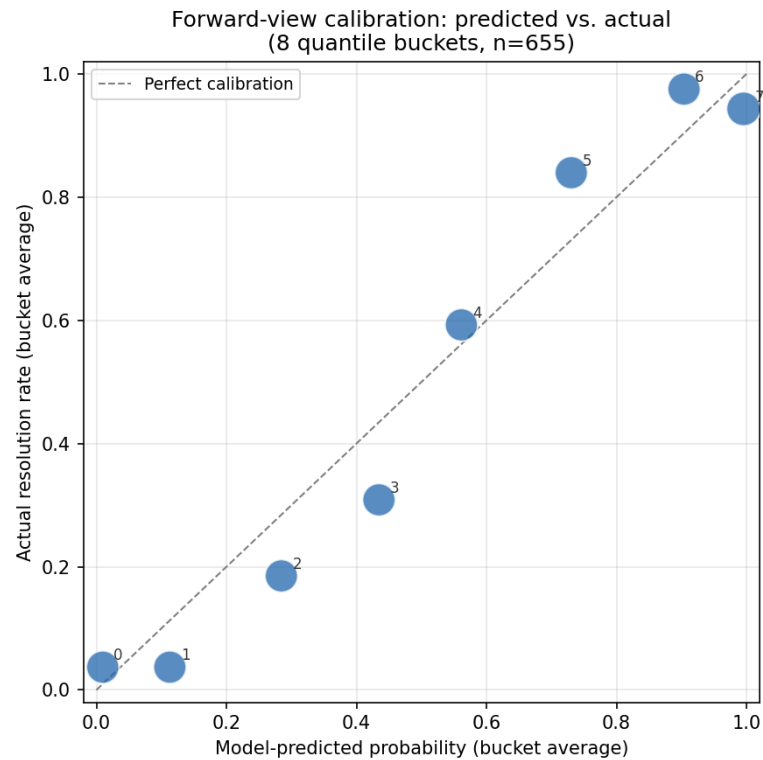


Figure 1: Predicted vs. actual resolution rate across 8 quantile buckets. Perfect calibration would place every point on the dashed diagonal. Marker size is proportional to bucket population.

Two things are visible in Figure 1. First, the ordering is almost perfectly monotonic – markets the model ranked as more likely to resolve YES genuinely did resolve YES more often, confirming the model carries real discriminative signal rather than noise. Second, the points do not sit exactly on the diagonal: an **S-shaped miscalibration** is visible, most clearly in Figure 2.

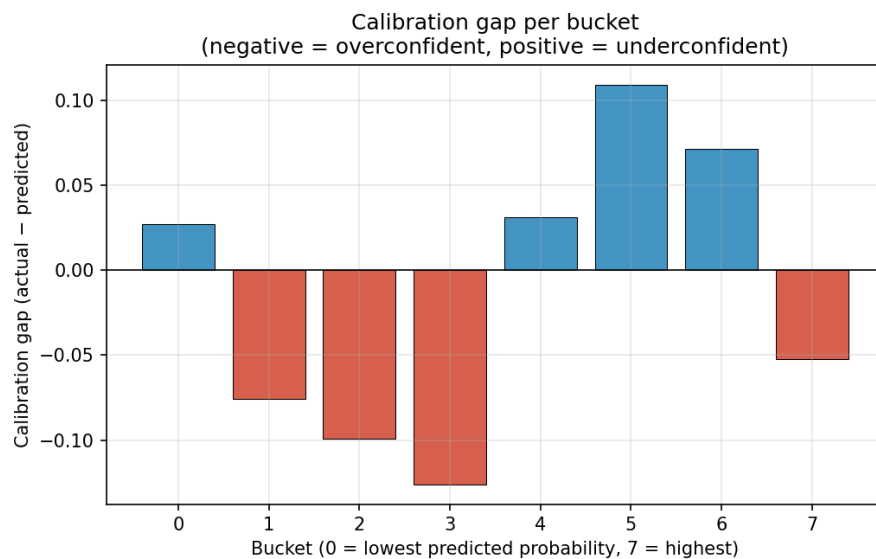


Figure 2: Calibration gap (actual – predicted) per bucket. Negative values indicate the model was overconfident in that probability range; positive values indicate underconfidence.

Buckets 1–3 (roughly the 11%–43% predicted range) are consistently **overconfident** by 7.6–12.6 percentage points; buckets 5–6 (73%–90% predicted) are consistently **underconfident** by 7–11 points. This pattern held across every checkpoint offset tested (Figure 3), and is consistent with the model’s principal known limitation: Deribit’s DVOL is a 30-day constant-maturity volatility index, used here as a proxy for volatility over horizons as short as two minutes. Black–Scholes assumes a flat volatility term structure; real markets do not always satisfy this, and the resulting mismatch plausibly explains a meaningful share of the observed miscalibration.

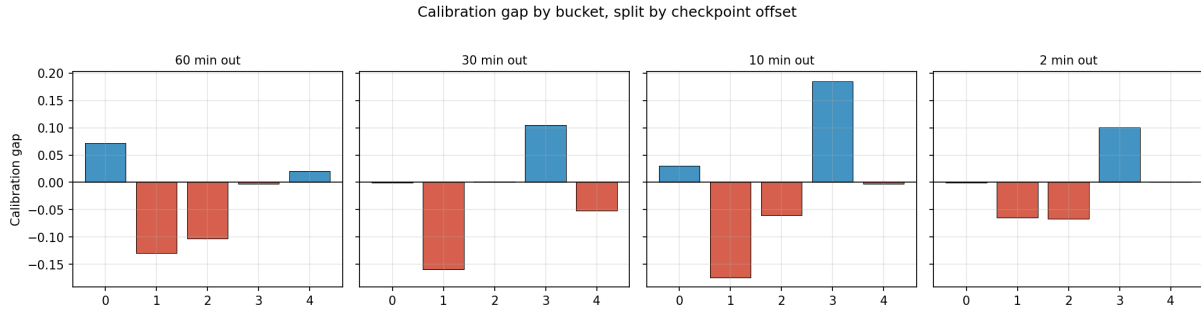


Figure 3: Calibration gap by bucket, split by time remaining to expiry at the checkpoint. The overconfidence bias in the low-probability range persists across every horizon; near-certain outcomes (buckets 0 and 4) become well-calibrated only in the final two minutes before close.

Notably, near-certain outcomes (the extreme buckets) become almost perfectly calibrated in the final two minutes before expiry – a gap of -0.0005 and $+0.0005$ respectively – consistent with both the model and the real Kalshi market converging on the truth as genuinely all the relevant information becomes available.

4.2 Inverse-View: The Volatility Gap Hypothesis Test

For each checkpoint, Kalshi’s midpoint price was inverted via the implied-volatility solver (Equation (4)) and compared directly against the Deribit DVOL fetched at that same moment. To avoid contaminating the test with the extreme numerical instability inherent to near-boundary probabilities (where a one-cent price change implies an enormous swing in backed-out volatility), checkpoints with $|p_{\text{mid}} - 0.5| > 0.45$ or an implied volatility exceeding 200% were excluded, leaving $n = 162$ usable observations.

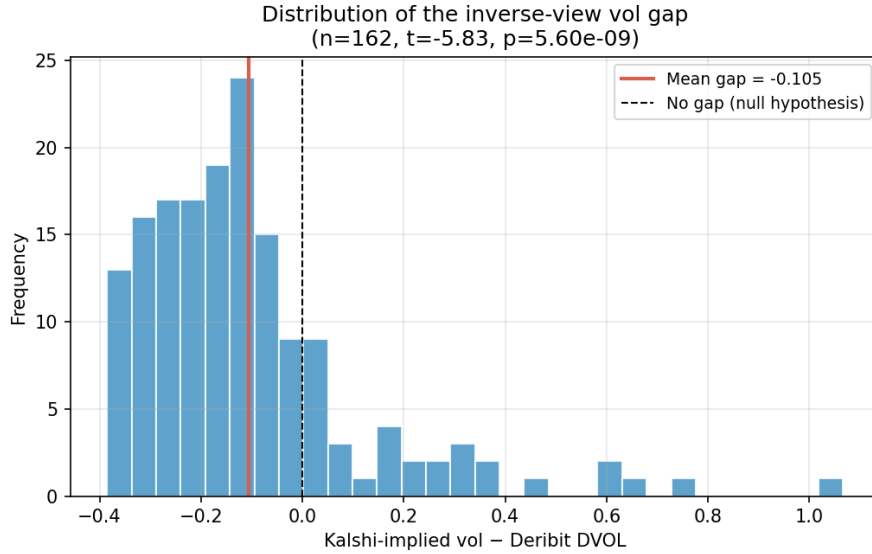


Figure 4: Distribution of the gap between Kalshi-implied and Deribit-implied (DVOL) volatility, restricted to non-boundary checkpoints.

Table 1: One-sample t -test results for the volatility gap

Statistic	Value
n	162
Mean gap $\bar{x}$	-0.1051
Sample std. dev. s	0.2294
Standard error	0.0180
t -statistic	-5.828
Two-tailed p -value	5.60×10^{-9}

The null hypothesis – that Kalshi-implied and Deribit-implied volatility are equal on average – is rejected overwhelmingly. Kalshi-implied volatility is, on average, **10.5 percentage points lower** than Deribit’s, a gap that would arise by chance roughly once in 178 million samples if the true difference were zero. This is a genuine, statistically robust finding: it does not, by itself, identify the mechanism (the term-structure explanation in Section 6 is a leading hypothesis, not a proven cause), but it does establish that the gap is real and systematic rather than sampling noise.

5 Simulated Trading Book

A simple trading rule was simulated against the same dataset: for every checkpoint where the model’s edge, $|p_{\text{model}} - p_{\text{Kalshi, mid}}|$, exceeded a threshold set to the average absolute calibration gap across the 8 buckets ($\theta = 0.0741$), a position was taken in the direction of the model’s implied edge, at the real ask price (accounting for the bid-ask spread), sized proportionally to the edge magnitude:

$$\text{size} = \text{base} \times \frac{|\text{edge}|}{\theta}$$

with realised P&L computed as $(\text{payout} - \text{entry price}) \times \text{size}$, where payout is \$1 if the position’s implied direction matched the real settlement outcome and \$0 otherwise.

Table 2: Trading book results under successively more disciplined rules

Rule	Trades	Total P&L	Avg. P&L/trade	Win rate
Naive (uncapped size)	294	+38.87	+0.132	24.1%
+ boundary exclusion	214	+36.83	+0.172	32.2%
+ size capped at 2	214	+22.32	+0.104	32.2%
+ size capped at 3	214	+27.38	+0.128	32.2%
+ size capped at 4	214	+30.86	+0.144	32.2%

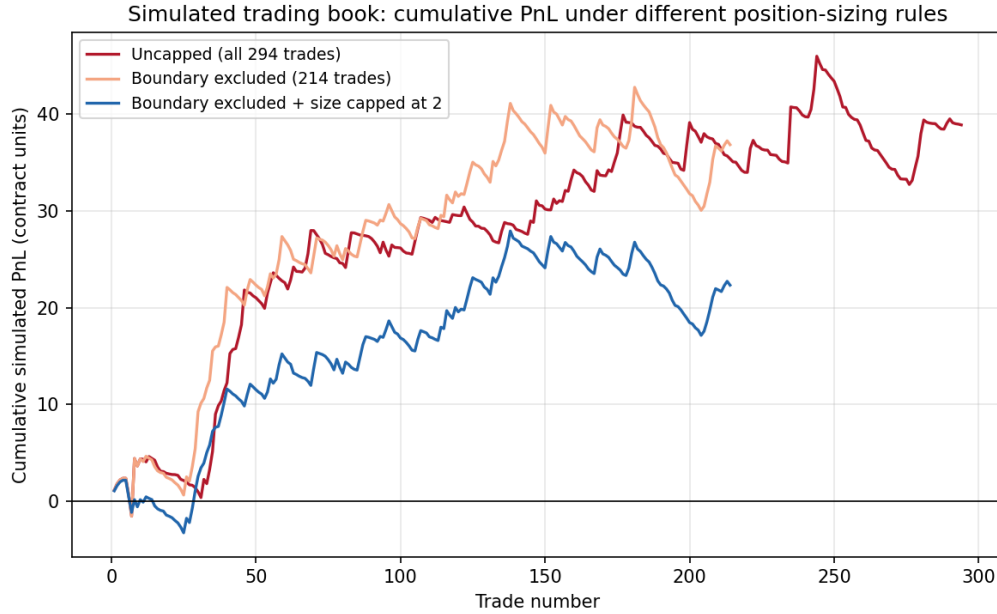


Figure 5: Cumulative simulated P&L under three progressively more disciplined rules.

The naive, uncapped rule is profitable in aggregate but structurally fragile: it wins only 24.1% of trades, and the top five trades alone account for **58.1%** of total profit – a small number of large, low-probability bets carrying almost the entire result. This is because position size scales directly with edge magnitude, and the largest edges are concentrated precisely at the extreme, near-boundary probabilities already identified as numerically unstable.

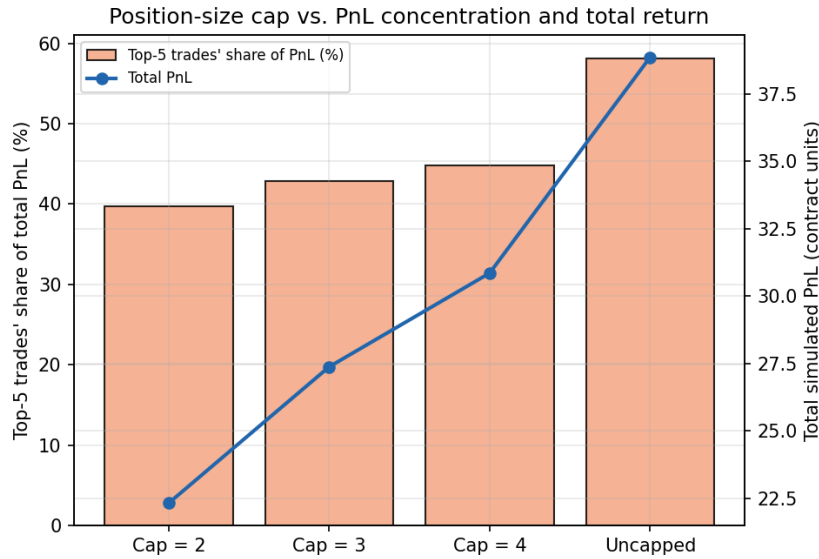


Figure 6: Effect of a position-size cap on both total P&L and profit concentration (top-5 trades' share of total P&L).

Excluding boundary probabilities alone lifts the win rate to 32.2% while barely denting total P&L, confirming that the excluded trades were not doing meaningful work. Capping position size further reduces concentration monotonically (58.1% $\rightarrow$ 39.7% at cap = 2) while total P&L declines more gently, and **remains clearly positive even at the tightest cap tested** (+22.32 over 214 trades, average +0.104 per trade). This is the central, honest conclusion of the trading simulation: the edge identified by the pricing model is real and survives a disciplined position-sizing rule, but the naive version of the strategy was closer to variance-harvesting on thin-tail bets than to a demonstrated systematic edge.

6 Limitations

- **Volatility term structure.** DVOL is a 30-day constant-maturity index used as a proxy for contracts expiring in minutes to an hour. Black–Scholes assumes flat volatility across all horizons, which real markets frequently violate. The most direct fix – using the nearest-listed Deribit option's own implied volatility rather than DVOL – was investigated but found infeasible for historical backtesting, since Deribit's live option chain has no equivalent historical endpoint; only DVOL supports arbitrary historical timestamp queries.
- **Sample scope.** All findings are drawn from a single asset (BTC), a single Kalshi series (KXBTCD), and a single 7-day window. The findings are internally consistent and statistically significant within that scope, but should not be generalised to other assets, series, or time periods without further validation.
- **Pseudo-replication.** Each hourly market contributes four checkpoints, which are correlated (they share a final outcome and much of their market context) rather than fully independent. The hypothesis test and trading simulation both treat all 655 observations at face value; a more conservative analysis restricted to one checkpoint per hour ($n \approx 164$) was examined and found directionally consistent, but was not used as the primary result.
- **No modelled transaction costs beyond the bid-ask spread.** Real Kalshi trading incurs exchange fees not modelled here.

7 Conclusion

This project set out to test, from first principles, whether statistical mispricing exists between Kalshi’s short-dated Bitcoin markets and Deribit’s options-implied volatility. The evidence supports a qualified yes: the pricing model built here shows genuine, monotonic discriminative power; the gap between Kalshi-implied and Deribit-implied volatility is statistically significant to an extremely high degree of confidence; and a disciplined simulated trading rule preserves a positive, non-trivial edge after correcting for the naive rule’s concentration risk. At the same time, the model exhibits a systematic, S-shaped calibration bias most plausibly explained by a volatility term-structure mismatch, and the trading result – while real – is modest and drawn from a single week of one asset. The project’s main technical achievement is the fully self-derived pricing and statistics pipeline: every piece of mathematics, from the normal CDF through the implied-volatility solver to the hypothesis test, was implemented and independently verified without reliance on external statistical libraries.